

Time-Series Agent system

A module-level construction view

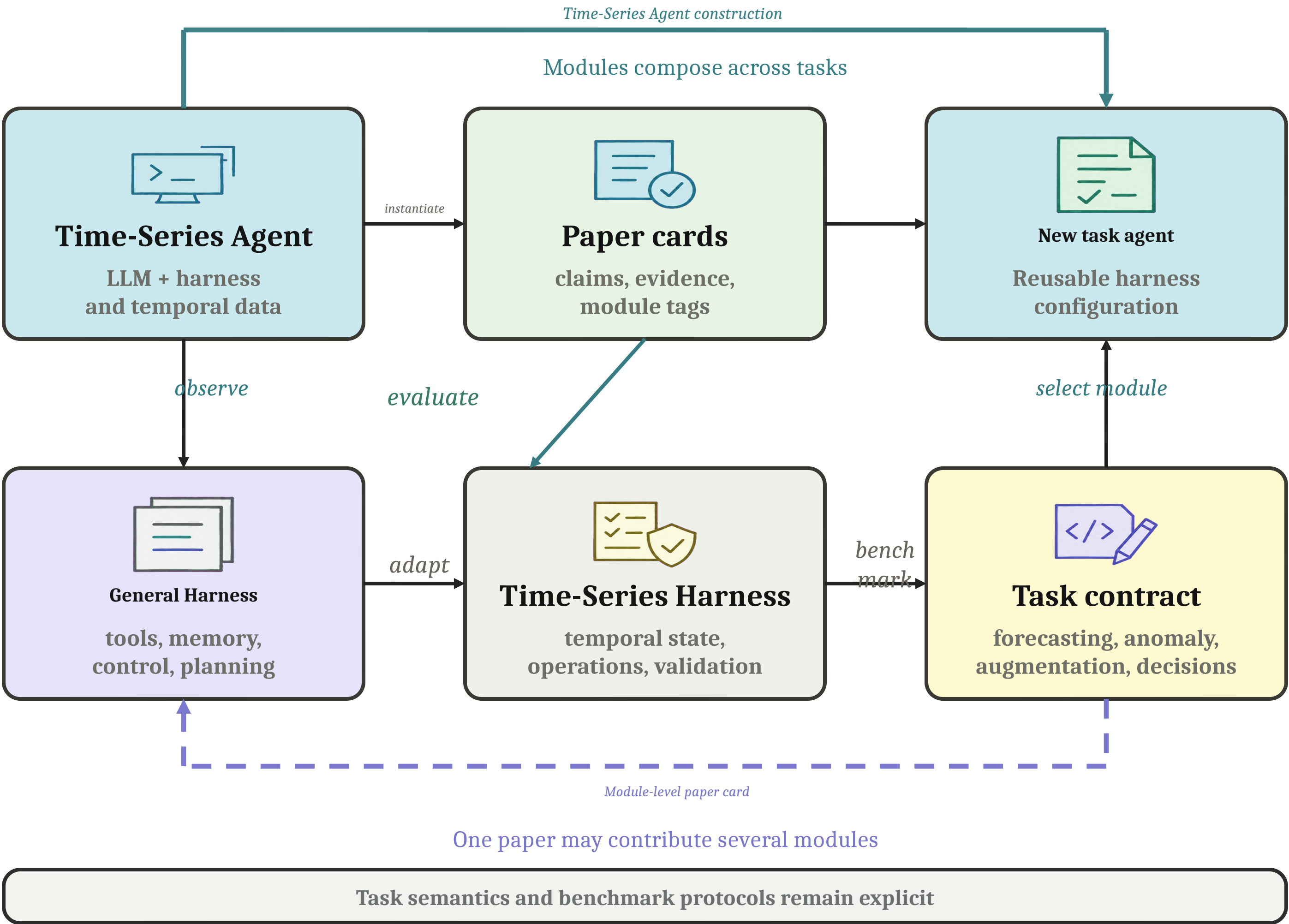


Figure 1. A module-level construction view of a Time-Series Agent. The LLM supplies language and reasoning; the General Harness provides reusable runtime mechanisms; the Time-Series Harness Module supplies temporal state, operations, grounding, and task contracts. A paper card records the claim, evidence locator, module layer, task family, and BibTeX entry.

Time-Series Agent

Reader's guide

Survey organization	Sections
§ 1 Introduction	System definition Construction questions
§ 2 What Is a Time-Series Agent?	Review scope Paper-card evidence
§ 3 LLM-Side Contributions	Temporal representation Alignment and reasoning LLM-side methods
§ 4 General Harness	Tools and memory Temporal state Control and execution Provenance and validation
§ 5 General Time-Series Harness	Time-series harnesses Task contracts Reusable modules Benchmark protocols
§ 6 Task-Specific Harness	Forecasting Anomaly diagnosis Augmentation Decision support
§ 7 Infrastructure and Benchmarks	ModernTSF Traces and episodes TimeSeriesGym Reproducible evaluation
§ 8 Open Problems and Case Studies	Paper-card database Open tasks Module-level counts Evidence gaps
§ 9 Conclusion	System map Open problems

Contribution layers in Time-Series Agents

Paper cards record module-level contributions across the LLM, Harness, temporal, task, and evaluation layers.

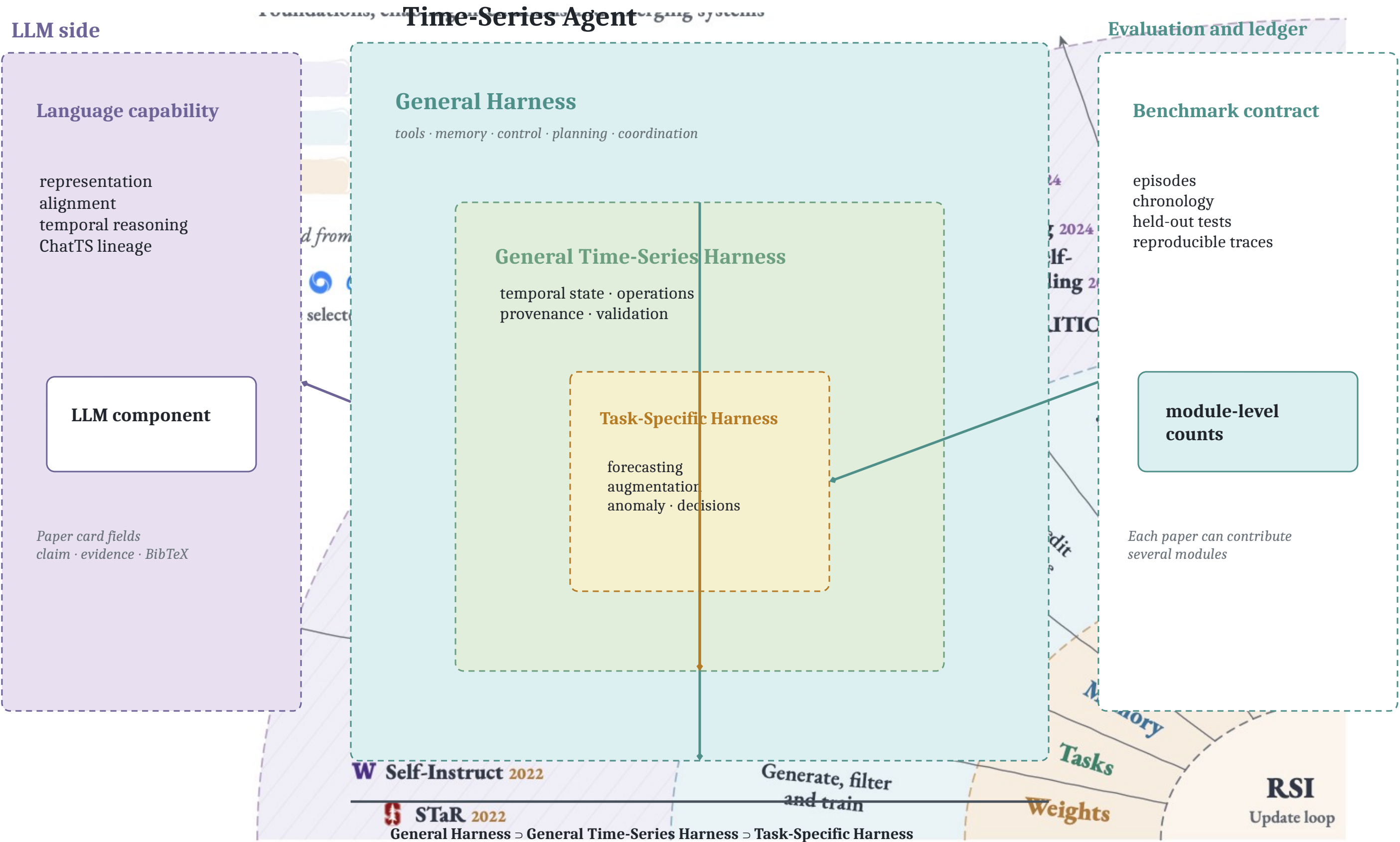


Figure 3. The survey records contributions at the module level. The LLM side is parallel to the Harness hierarchy; temporal modules specialize the reusable runtime, and task contracts instantiate the finest layer.

Research routes in Time-Series Agent systems

Representative module contributions by first public year | LLM and Harness layers

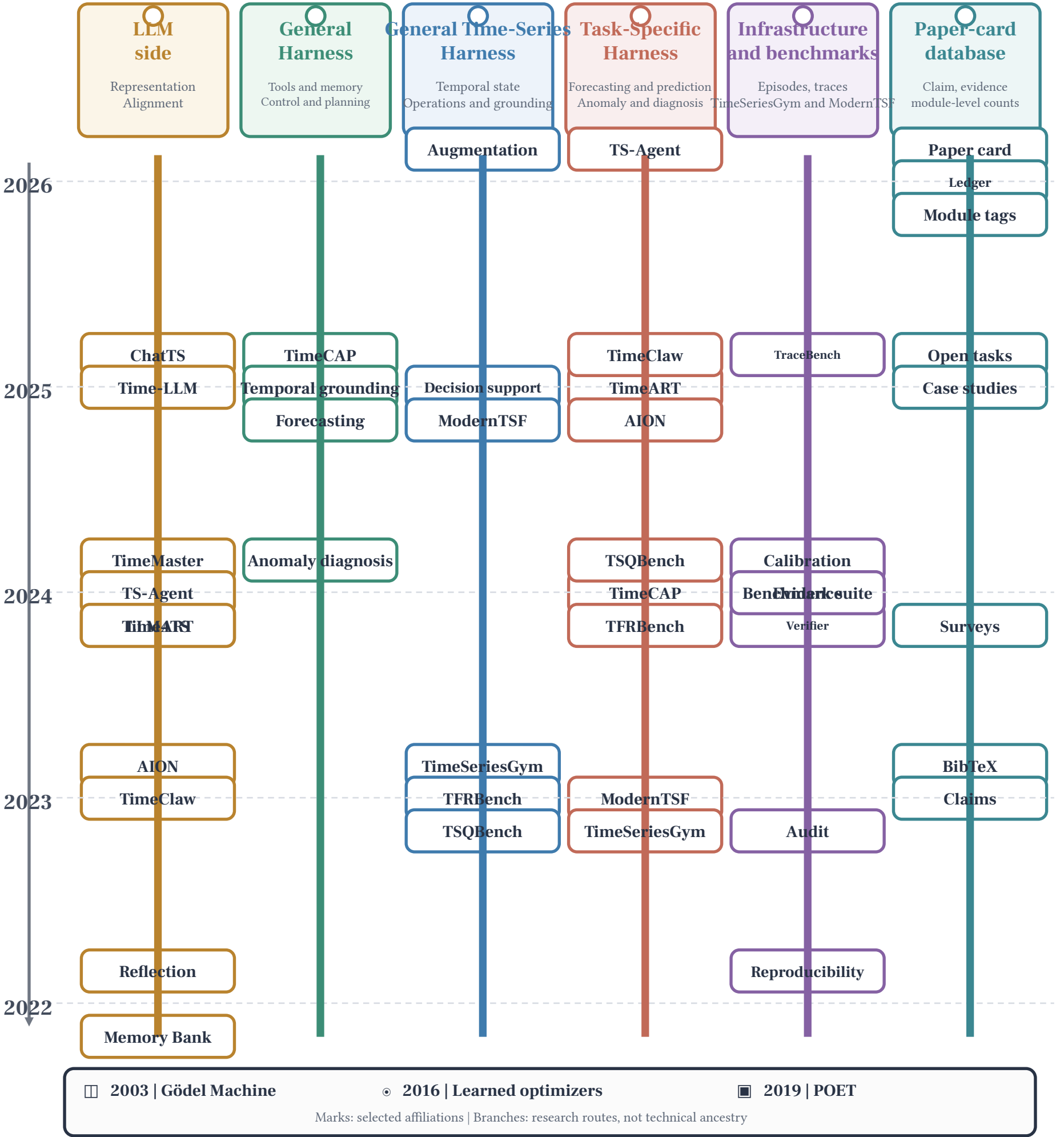


Figure 4. Research routes toward recursive self-improvement. Forty-five representative method papers are grouped into six research routes, with three historical anchors shown separately.

Year bands use the first public dates in the checked source records; spacing within each band is adjusted for legibility. Branch lines denote research families.

Technical ancestry, cross-route transfer, and sustained recursive gains require separate evidence. The selection reflects the literature search through September 12, 2026.

Logos identify one selected author or work-done affiliation per study; “Ind.” denotes independent researchers. Affiliation and logo sources are documented separately.

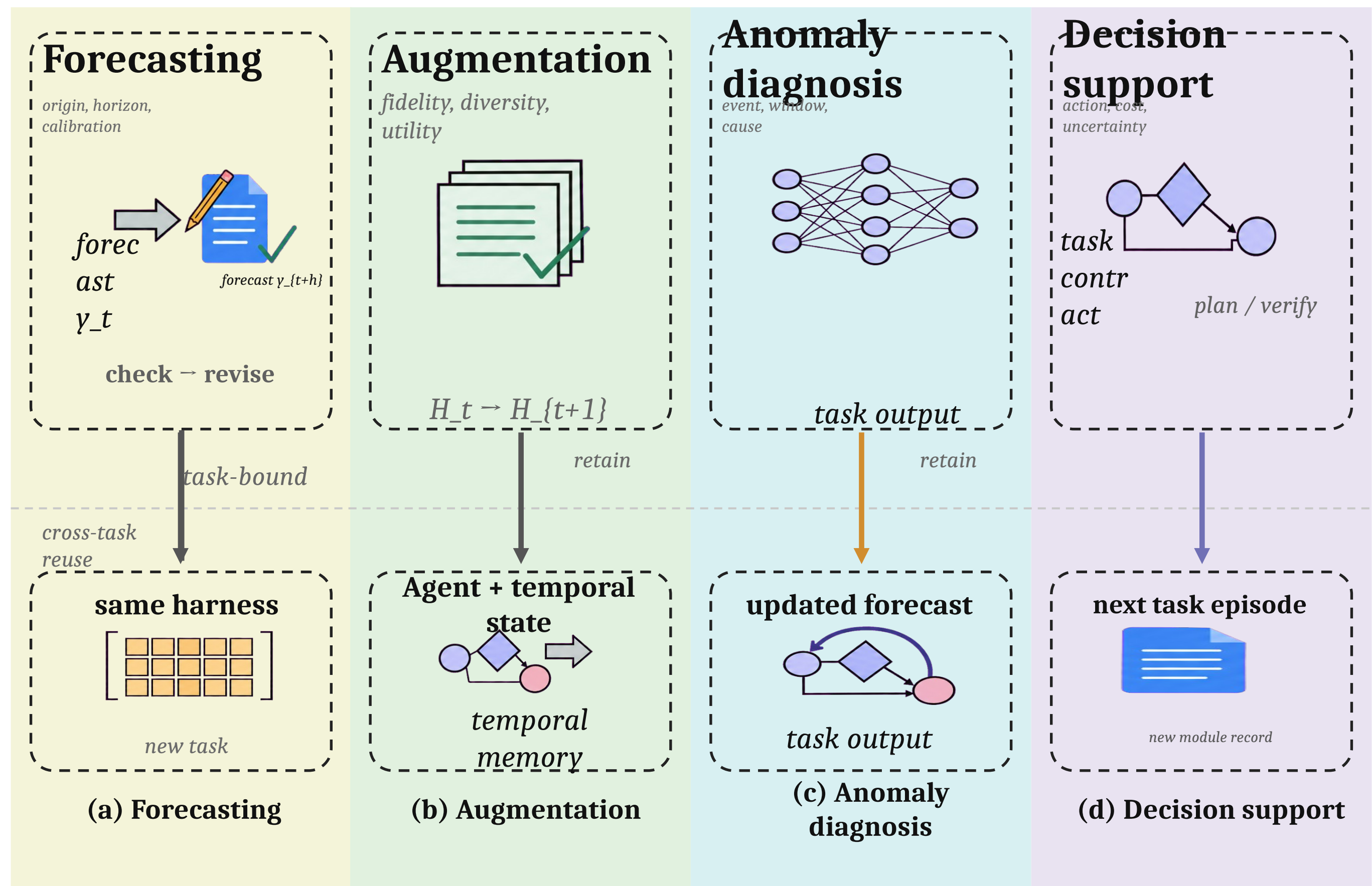


Figure 5. Four task contracts instantiated by Time-Series Harness Modules. The same general runtime can support distinct temporal outputs when the interface, memory, tools, control, planning, optimization, and coordination obligations are specialized.

Forecasting binds origin, horizon, and calibration. Augmentation binds fidelity, diversity, and utility. Anomaly diagnosis binds event windows and causes. Decision support binds actions, costs, and uncertainty.

A paper may instantiate more than one contract. The paper-card ledger records each module independently so that a task-specific contribution does not occupy a coarser layer.

The dashed horizontal rule separates reusable harness mechanisms from task-bound contracts.

Colored arrows show the output contract carried by each task family.

The visual vocabulary is adapted from the supplied RSI template; the content follows our four task families.